The Gaussian Copula Prior

Correlated priors for ReplayBG twinning

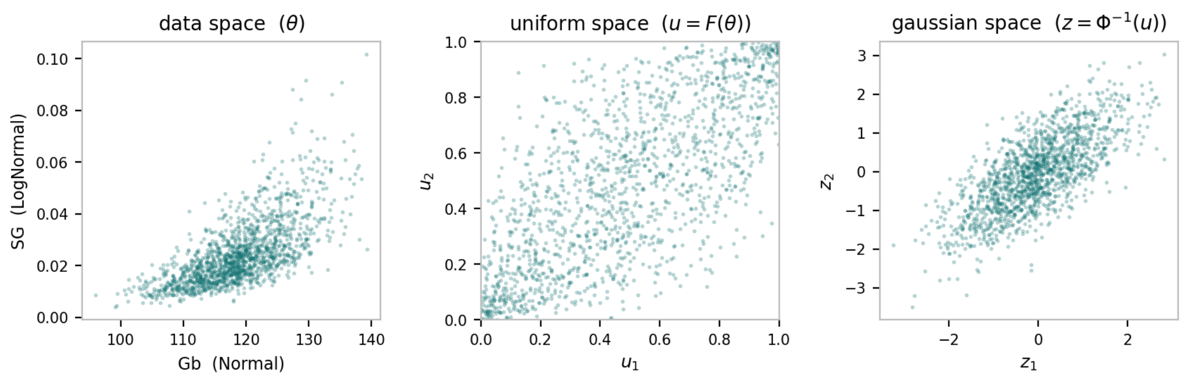
ReplayBG estimates the unknown model parameters θ by maximizing the log-posterior, log-prior + log-likelihood. This note explains, from the ground up, how a **Gaussian copula** lets us impose a prior correlation between parameters while keeping each parameter's marginal distribution exactly as declared — starting with what a copula is and why anyone would want one.

1. A short introduction to copulas

The word *copula* is Latin for a 'link' or 'bond', and that is precisely its job: a copula is the mathematical object that links several one-dimensional distributions into a single joint distribution. The concept was introduced by Abe Sklar in 1959 and has since become a standard tool in quantitative finance, insurance, hydrology, reliability engineering and, increasingly, statistical modelling and machine learning.

Its power comes from a clean separation of concerns. Any joint distribution answers two distinct questions: **(a) what does each variable look like on its own?** — its *marginals*, the individual shapes — and **(b) how do the variables move together?** — their *dependence structure*. A copula isolates the second question completely. It is built by stripping every variable of its individual shape (mapping each one to a featureless Uniform(0,1) scale through its own CDF), so that whatever remains is *pure dependence* and nothing else. Formally, a copula is simply a joint cumulative distribution function on the unit cube whose marginals are all uniform.

This separation is exactly what we need. We want to keep each parameter's declared marginal untouched (Gb stays Normal, SG stays LogNormal) yet still make the parameters dependent. Copulas let us pick the marginals and the dependence *independently* and then glue them back together. The figure below shows the idea concretely: the same 1500 samples viewed in three spaces. The marginals are arbitrary (a Normal Gb against a skewed LogNormal SG), yet a controlled correlation has been imposed — visible as the tilted cloud once both variables are mapped to the neutral uniform and Gaussian scales.



Left: the raw parameter cloud, each axis keeping its own marginal shape. Middle: after the probability-integral transform, the dependence shown on uniform margins — this is the copula. Right: after the inverse normal transform, the dependence becomes a tilted Gaussian ellipse controlled by a single correlation.

Many copula families exist — Gaussian, Student-*t*, and the Archimedean family (Clayton, Gumbel, Frank), among others — each encoding a different flavour of dependence. This note uses the **Gaussian copula**: the dependence structure borrowed from a multivariate normal, which is the most natural first choice and the one implemented in ReplayBG.

2. Why a prior correlation matters

Twinning is MAP estimation: it maximizes log-likelihood + log-prior. The prior encodes what we believe about the parameters before seeing this subject's CGM data. With an **independent** prior we assert that learning one parameter tells us nothing about another — a strong claim, since physiology is full of couplings (insulin sensitivity and glucose effectiveness, the gastro-intestinal absorption constants, and so on, tend to co-vary across a population). A correlated prior says that regions of parameter space where two parameters move together are *a priori* more plausible than regions where they move oppositely. When the likelihood is weak (short or noisy data) that coupling is what keeps the estimate on the right ridge instead of letting parameters wander independently.

3. Why you cannot just 'add' correlation

A multivariate distribution is **not** pinned down by its marginals plus a correlation number. Given marginals there are infinitely many joint distributions sharing them; the independent product is only one. And there is no closed-form joint density with a Normal margin *and* a LogNormal margin — the bivariate normal forces both margins to be normal, so it is simply unavailable here. We need a construction that keeps arbitrary, mismatched marginals exactly while injecting a controllable dependence. That construction is the copula.

4. Sklar's theorem (the foundation)

Sklar's theorem is the bridge. For any joint CDF H with marginals $F_1,\dots,F_k$ there exists a copula C with

$$H(\theta_1, \dots, \theta_k) = C(F_1(\theta_1), \dots, F_k(\theta_k)),$$

and when the marginals are continuous, C is **unique**. The converse holds too: take any marginals and any copula, glue them with this formula, and the result is a valid joint distribution with exactly those marginals. Differentiating gives the density we actually use,

$$h(\theta) = \left[\prod_i f_i(\theta_i)\right] \cdot c(F_1(\theta_1), \dots, F_k(\theta_k)),$$

where c is the copula density. The bracket is your marginals, untouched; c is the correction that introduces dependence. If $c \equiv 1$ (the independence copula) you are back to the plain product.

5. The probability integral transform

The mechanism that maps any marginal onto the neutral uniform scale is the probability integral transform: if X has continuous CDF F, then U = F(X) is Uniform(0,1). The one-line proof:

$$P(U \le u) = P(F(X) \le u) = P\left(X \le F^{-1}(u)\right) = F\left(F^{-1}(u)\right) = u,$$

which is the Uniform(0,1) CDF. So every continuous variable becomes uniform after passing through its own CDF, and the inverse transform $X = F^{-1}(U)$ goes back. This is why each distribution needed a `.cdf()` method: $u_i = F_i(\theta_i)$ is the transform that puts a Normal margin and a LogNormal margin on common ground.

6. Building the Gaussian copula

A copula is a choice; the Gaussian copula is the dependence of a multivariate normal transplanted onto your marginals, in three steps. **(a) To uniform** via the transform of §5: $u_i = F_i(\theta_i)$. **(b) To standard normal** through the inverse standard-normal CDF, $z_i = \Phi^{-1}(u_i)$, so each z_i is marginally N(0,1). **(c) Couple** by declaring

$$z = (z_1, \ldots, z_k) \ \sim \ \mathcal{N}(0, R),$$

with R a correlation matrix (unit diagonal). All dependence lives in R; the margins stay N(0,1) because the off-diagonal of R does not change individual variances. The copula density is the joint normal density at z divided by what independent normals would give — the $(2\pi)^{-k/2}$ normalizers cancel (the whole point of a copula: no marginal content is left):

$$c_R(u) = \frac{\varphi_R(z)}{\prod_i \varphi(z_i)} = |R|^{-1/2} \exp\left(-\tfrac{1}{2}\, z^T (R^{-1} - I)\, z\right), \quad z_i = \Phi^{-1}(u_i).$$

A useful sanity check: if every marginal is itself Normal, the Gaussian copula with Normal margins reproduces an ordinary multivariate Normal exactly. The copula is the generalization that keeps that dependence while letting the margins be anything.

7. The log-prior the twinner evaluates

Combining the copula density with the marginals and taking logs gives the log-prior — the first term is the unchanged sum of marginals, the rest is the copula correction:

$$\log p(\theta) = \sum_i \log f_i(\theta_i) \ - \ \tfrac{1}{2} \log|R| \ - \ \tfrac{1}{2}\, z^T (R^{-1} - I)\, z, \quad z_i = \Phi^{-1}(F_i(\theta_i)).$$

8. Mapping to the code

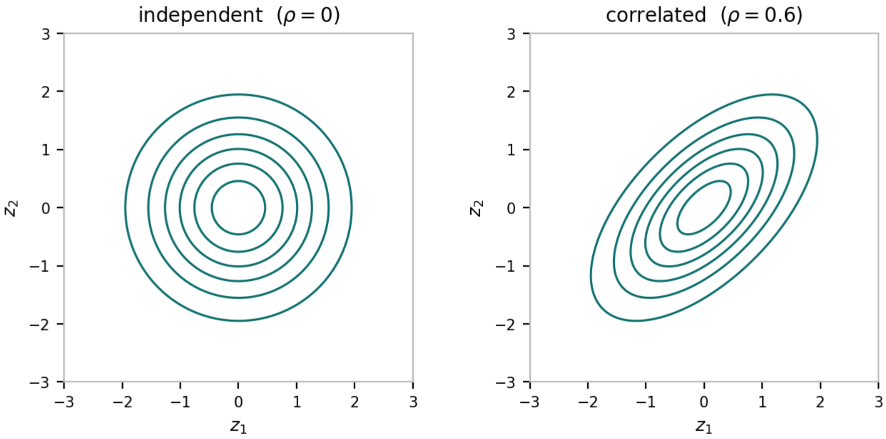
That correction is exactly the term added inside `_log_prior`:

```
lp += -0.5 * correlation_structure['logdet_R'] \
      - 0.5 * float(z @ correlation_structure['R_inv_minus_I'] @ z)
```

`logdet_R` = log|R| and `R_inv_minus_I` = R^{-1}–I are precomputed once in `_build_correlation_structure` (via `slogdet` and `inv`), so each objective evaluation is a single quadratic form $z^T M z$ — $O(k^2)$ in the number of correlated parameters (typically 2–4). The z_i come from `ndtri(cdf(·))`, clipped to $[10^{-12}, 1–10^{-12}]$ so Φ^{-1} stays finite at the boundaries.

9. Geometric intuition

Read the log-prior as an energy landscape. The marginal term alone gives axis-aligned contours. The copula correction adds the quadratic form, which tilts and stretches those contours in the latent z-space: with $\rho > 0$ the penalty is smallest along the diagonal where both z's share a sign (both parameters jointly high or jointly low) and largest on the anti-diagonal. The quantity $z^T R^{-1} z$ is a Mahalanobis distance, so off-ridge combinations sit far away and become unlikely; the optimizer then slides along the favoured ridge instead of treating the parameters as a free rectangle.



Latent-Gaussian prior contours. Independent priors give circular (axis-aligned) contours; a positive correlation tilts them into ellipses along the joint-high / joint-low diagonal.

10. How to choose R

From a population of past twins. R is a correlation in the latent Gaussian space, so estimate it robustly through a *rank* correlation, which is invariant to the marginal shapes. From Spearman's ρ_S or Kendall's τ:

$$R_{ij} = 2\sin\left(\tfrac{\pi}{6}\rho_S\right) \qquad \text{or} \qquad R_{ij} = \sin\left(\tfrac{\pi}{2}\tau\right).$$

From domain knowledge. Pick a modest $|\rho|$ (say 0.3–0.5) to express a believed coupling without over-committing, and check sensitivity. **Validity:** R must be symmetric positive-definite. With more than two correlated parameters the pairwise values are not free — e.g. $\rho_{12} = \rho_{13} = 0.9$, $\rho_{23} = -0.9$ is not a valid correlation matrix. The `cholesky(R)` guard turns that into a clear error rather than a silent NaN.

11. The attenuation caveat

$\rho = R_{ij}$ is the correlation between the Gaussian scores z_i, z_j — not the Pearson correlation of θ_i, θ_j. The induced correlation on the raw parameters follows Hoeffding's identity,

$$\mathrm{Cov}(\theta_i, \theta_j) = \iint \left[H_{ij}(s,t) - F_i(s)F_j(t)\right] ds\, dt,$$

which depends on the marginals. The map from ρ to Pearson(θ) is monotone and passes through 0, but for a strongly skewed margin (the LogNormal SG) the Pearson correlation on the θ scale comes out *smaller* than ρ. So to reproduce a measured Pearson correlation exactly, do not type it directly — go through a rank correlation (§10), which is invariant. For 'a sensible positive coupling', typing ρ directly is fine.

12. Limitations and where to go beyond

No tail dependence. The Gaussian copula has zero coefficient of tail dependence: even at $\rho = 0.95$, conditioning on one parameter being extreme does not make the other reliably extreme ($\lambda = \lim_{q\to1} P(U_2 > q \mid U_1 > q) = 0$ for $\rho < 1$). This very property is what made the Gaussian copula infamous in pre-2008 credit-risk models. If joint-extreme behaviour matters, a Student-*t* copula adds symmetric tail dependence and slots into the same z-space machinery.

Symmetric dependence only. Gaussian and t copulas are radially symmetric; ‘correlated in the low range, independent in the high’ needs Archimedean copulas (Clayton, Gumbel) or vine copulas. **MAP vs full Bayes.** The correction shifts the mode the optimizer finds, which is what twinning wants; the same log-prior serves unchanged if ReplayBG ever moves to posterior sampling. **Initial sampling.** Multi-start guesses are deliberately left independent; drawing starts from the joint copula too would additionally need an inverse-CDF (`ppf`) per distribution — a clean extension that changes only where the optimizer starts, not the objective.

API: `rbg.twin(..., correlations={('Gb','SG'): 0.4})` — pairwise correlations; unlisted pairs default to 0; the joint prior is the Gaussian copula described here.